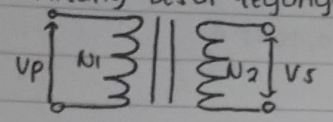
Nama : Ailsa Rahma
NIM : 241331002

IATTE

Latsol Indikator & Transformator

1. a). Hitung besar tegangan pada terminal sekundernya (Vs)



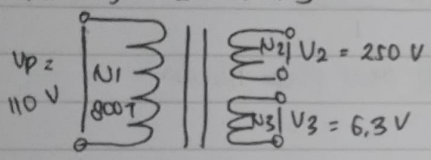
Vp = 220V 50Hz N2 = 600 lilitan
N1 = 1200 lilitan

↳ $\frac{Vp}{Vs} = \frac{Np}{Ns} \rightarrow \frac{220V}{Vs} = \frac{1200}{600} \rightarrow Vs = \frac{220 \cdot 600}{1200} = 110$ watt

b). Hitung lagi Vs bila sekundernya ditambah sebanyak 80 lilitan

↳ $\frac{Vp}{Vs} = \frac{Np}{Ns} \rightarrow \frac{220V}{Vs} = \frac{1200}{600+80} \rightarrow Vs = \frac{220 \cdot 680}{1200} = 124,677$ watt

2. Hitung N_2 dan N_3 bila diinginkan tegangan outputnya seperti :



↳ $\frac{\frac{1}{2} Vp}{V_2} = \frac{\frac{1}{2} Vp}{N_2}$

$\frac{\frac{1}{2} \cdot 110}{250} = \frac{\frac{1}{2} 800}{N_2}$

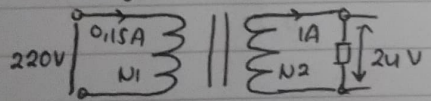
$N_2 = \frac{250 \cdot 400}{55} = 1818,9$ T

$\frac{\frac{1}{2} Vp}{V_3} = \frac{\frac{1}{2} Np}{N_3}$

$\frac{\frac{1}{2} 110}{6,3} = \frac{\frac{1}{2} 800}{N_3}$

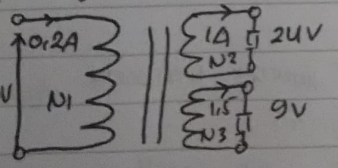
$N_3 = \frac{400 \cdot 6,3}{55} = 45,81$ T

3. Hitunglah effisiensi (n)



↳ $n = \frac{Vs \cdot Is}{Vp \cdot Ip} \times 100\% = \frac{24.1}{220 \cdot 0,15} \times 100\% = 72,72\%$

4. Hitunglah effisiens (n)



↳ $P_S = P_2 + P_3$
$= (V_{S2} \cdot I_{S2}) + (V_{S3} \cdot I_{S3})$
$= (24.1) + (9.(1,5) = 37,5$ watt

↳ $P_P = V_{P1} \cdot I_{P1}$
$= 220 \cdot 0,2 = 44$ Volt

$n = \frac{P_S}{P_P} \times 100\% = \frac{37,5}{44} \times 100\% = 85,227\%$

5. Hitung arus primer I_1, bila effisiensi (η) dari transformer sebesar 80%

↳ $\eta = \frac{V_s \cdot I_s}{V_p \cdot I_p} \times 100\%$

$80\% = \frac{12.3}{220 \cdot I_1} \times 100\%$

$\frac{4}{5} = \frac{36}{220\, I_1}$

$220\, I_1 = \frac{5 \cdot 36}{4} = 45$

$I_1 = 0,2045\ A$

6. Transformator daya dengan perbandingan 1 : 8 pada terminal primer dipasang ke Vs bolak balik dengan tegangan 220V 50Hz

a). berapa besar Frekuensi

↳ $F_s = 50\ Hz$

b). berapa besar tegangan sekunder

↳ $\frac{V_p}{V_s} = \frac{N_p}{N_s} \rightarrow \frac{220}{V_s} = \frac{1}{8} \rightarrow V_s \frac{220 \cdot 8}{1} = 1760\ V$

c). bila dipasang beban 10kΩ, berapa arus pada lilitan

↳ $I_s = \frac{V_s}{R} = \frac{1760}{10k} = 0,176\ A$

d). berapa arus pada primer bila effesiensi 100%

↳ $100\% = \frac{1760 \cdot 0,176}{220 \cdot I_p} \times 100\%$ $\quad I_p = 1,408\ A$

7. Hitung nilai induktansi

a). inti udara, 20 lilitan, luas inti 3,14 cm2, panjang liliton 25 cm

↳ $L = \mu_r \cdot \mu_0 \cdot \frac{N^2}{P} \cdot A = 1 \cdot 1,26 \times 10^{-6} \cdot \frac{20^2}{0,25} = 0,000314$

b). $\mu_r = 5000$

↳ $5000 \cdot 1,26 \times 10^{-6} \cdot \frac{20^2}{0,25} = 0,000314$

c). Hitung nilai resistansi bila digunakan kawat tembaga dgn $\phi = 0,3mm$

↳ $A = \pi \cdot d = 3,14 \cdot 0,3 = 0,942\ mm$

$R = \rho \frac{l}{A} = 1,68 \times 10^{-8} \cdot \frac{0,25}{0,942} = 4,46 \times 10^{-9}\ \Omega$

d). Hitung besar faktor kualitas bila bekerja di $F = 1MHz$

↳ $X_L = 2\pi f L$

$= 2\pi \cdot 1 \times 10^6 \cdot 6,34 \times 10^{-7} = 3,981 \times 10^{-2}\ H$

$Q = \frac{X_L}{R} = \frac{3,981 \times 10^{-2}}{4,46 \times 10^{-9}} = 8,92 \times 10^{-6}\ C$